UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (10 MARKS)

Answer ALL questions.

1. What are parallel arrays?
   A. Arrays with no index
   B. Arrays whose elements at the same index are related
   C. Arrays that hold only characters
   D. Arrays stored in different programs

2. What is displayed by the following code?

```
for (int i = 1; i <= 4; i++)
    cout << i * 2;
```

   A. 246
   B. 1357
   C. 1234
   D. 2468

3. Which declaration creates a floating-point variable named price?
   A. bool price;
   B. char price;
   C. float price;
   D. int price;

4. Which part of a for loop is executed after each iteration?
   A. The initialisation
   B. The condition only
   C. The update expression
   D. The loop body twice

5. A computer system's hardware includes which of the following?
   i. input devices
   ii. output devices
   iii. central processing unit
   iv. main memory
   v. secondary storage
   A. i, ii, iii, v
   B. i, iii, iv
   C. i, ii, iii, iv
   D. i, ii, iii, iv, v

6. Which data type stores a whole number?
   A. string
   B. float
   C. char
   D. int

7. Which operator sends data to cout?
   A. ==
   B. =
   C. <<

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D. >>

8. Which phase of the program development life cycle defines what the program must do?
   A. Coding
   B. Testing
   C. Analysis
   D. Maintenance

9. Which of the following are valid identifiers?
   i. total_score
   ii. 2total
   iii. _count
   iv. int
   A. iii and iv
   B. ii and iv
   C. i and ii
   D. i and iii

10. How many times does a while loop execute when its condition is false initially?
   A. One
   B. Until the user stops it
   C. Two
   D. Zero

PART B (60 MARKS)

QUESTION 1

This question tests programming concepts and error terminology.

a) Define compiler and interpreter.

(2 marks)

b) State THREE differences between a compiler and an interpreter.

(3 marks)

QUESTION 2

A program reads a student's score from 0 to 100 and displays the grade shown in Table 1.

Table 1. Grade Scale

Score	Grade
80-100	A
60-79	B
50-59	C
40-49	D
0-39	F

a) Draw a flowchart for the grade classification.

(6 marks)

b) Write pseudocode for the algorithm and convert it into a complete C++ program.

(7 marks)

QUESTION 3

This question tests operator precedence and assignment.

a) Evaluate the following expressions using the declarations in the code block.

```
int a = 12, b = 5;
double c = 2.5;
```

(5 marks)

b) Write C++ statements to declare two integers a and b, read their values, and display their sum, difference, product and remainder.

(6 marks)

QUESTION 4

Study the following array program segment:

```
int a[5] = {2, 4, 6, 8, 10};
int s = 0;
for (int i = 0; i < 5; i++) {
    s += a[i];
}
cout << s;
```

a) Trace the program and state the output.

(6 marks)

b) Rewrite the segment using a do...while loop that sums only values greater than 5.

(9 marks)

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QUESTION 5

A sports club records the sport names and medal counts in Table 1.

a) Declare two parallel arrays for the five sports in Table 1 and initialise them with those values.

Table 2. Sport and Medal Records

Sport	Medals
Swimming	8
Badminton	12
Athletics	7
Cycling	5
Hockey	9

(6 marks)

b) Write a program segment to display each sport with its medals, then calculate the total and average medals.

(10 marks)

PART C (30 MARKS)

QUESTION 1

A cafe sells the items shown in Table 1. A 10% discount is given when the subtotal is at least RM50.

Table 3. Cafe Menu

Item	Price (RM)
Coffee	8.50
Tea	6.00
Juice	9.50

Write a complete C++ program that reads the item code and quantity, calculates the subtotal and discount, and displays a formatted receipt with the item name, quantity, subtotal, discount and amount to pay.

(15 marks)

QUESTION 2

A bookstore records up to five books. Table 1 shows an example of the fields.

Table 4. Book Record Fields

Title	Price (RM)	Quantity
Book A	45.00	3
Book B	30.00	5
Book C	55.00	2

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

END OF QUESTION PAPER